



Vectors - Worksheet 1

Chapter 10: Vectors. Answer all questions. Show your working clearly. Total: 104 marks. Time: 2 hours 10 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are five sections, one for each topic in Chapter 10. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- **i**, **j** and **k** are unit vectors in the x , y and z directions. Diagrams are not drawn to scale.
- Give non-exact answers to 3 significant figures, or angles to 1 decimal place.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. **[3]**.

Section	Marks available	Marks scored
A Vectors in 2D: magnitude, direction, unit vectors	18	
B Position vectors and dividing a line in a given ratio	20	
C Vector geometry: collinearity, parallelism, shape properties	20	
D Vectors in 3D and the scalar product	21	
E Vectors in mechanics	20	
Roar-level challenge	5	
Total	104	

Section A Vectors in 2D: magnitude, direction, unit vectors

18 marks

1. Given $\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$, find $|\mathbf{v}|$ and the unit vector in the direction of $\mathbf{v}$. [2]

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2. Given $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j}$, find $3\mathbf{a} - 2\mathbf{b}$ and $|3\mathbf{a} - 2\mathbf{b}|$. [2]

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3. Find the angle that the vector $4\mathbf{i} - 3\mathbf{j}$ makes with the positive x -axis, and write down the unit vector in the same direction. [3]

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4. (a) The vector $\mathbf{p} = k\mathbf{i} + 6\mathbf{j}$ has magnitude 10. Find the possible values of k . [4]
- (b) Find the vector of magnitude 15 in the same direction as $3\mathbf{i} + 4\mathbf{j}$.

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5. Two forces, $\mathbf{F}_1 = (4\mathbf{i} - 5\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + 3\mathbf{j})$ N, act on a particle, where p is a constant. The resultant force $\mathbf{R}$ is parallel to the vector $3\mathbf{i} - \mathbf{j}$.

(a) Find the value of p . [3]

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(b) Find the exact magnitude of $\mathbf{R}$. [2]

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(c) Find the angle between $\mathbf{R}$ and the vector $\mathbf{i}$. [2]

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(Total for Question 5 is 7 marks)



Section B Position vectors and dividing a line in a given ratio

20 marks

6. The points A and B have coordinates $(2, -1)$ and $(6, 5)$. [2]

Find $\vec{AB}$ in terms of $\mathbf{i}$ and $\mathbf{j}$, and the exact length of AB .

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7. The points A and B have position vectors $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = 9\mathbf{i} + 10\mathbf{j}$. M is the midpoint of AB . [3]

Find the position vector of M and the exact distance AM .

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8. The points A and B have position vectors $\mathbf{a} = -2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 6\mathbf{i} - 3\mathbf{j}$. [3]

The point P lies on AB such that $AP : PB = 1 : 3$. Find the position vector of P .

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9. Relative to an origin O , $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

[4]

M is the midpoint of AB , and N is the point on OB such that $ON : NB = 2 : 1$.

Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, (a) $\overrightarrow{AN}$, (b) $\overrightarrow{MN}$.

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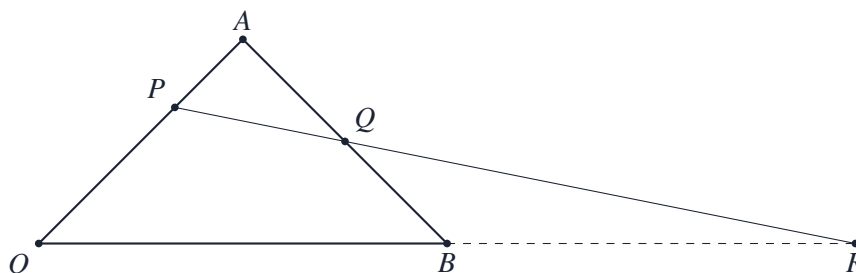
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10. In triangle OAB , $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.

The point P lies on OA such that $OP : PA = 2 : 1$, and Q is the midpoint of AB .



(a) Find $\vec{PQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

[2]

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(b) The point R lies on OB extended, with $\vec{OR} = k\mathbf{b}$. Given that P , Q and R are collinear, find the value of k .

[4]

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(c) Find the ratio $PQ : QR$.

[2]

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(Total for Question 10 is 8 marks)



Section C Vector geometry: collinearity, parallelism, shape properties

20 marks

11. Show that the vectors $\mathbf{u} = 2\mathbf{i} - 6\mathbf{j}$ and $\mathbf{w} = -3\mathbf{i} + 9\mathbf{j}$ are parallel. [2]

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12. The points A , B and C have position vectors $\mathbf{i} + 2\mathbf{j}$, $4\mathbf{i} + 3\mathbf{j}$ and $10\mathbf{i} + 5\mathbf{j}$. [3]
Prove that A , B and C are collinear.

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13. The points $A(1, 2)$, $B(3, k)$ and $C(7, 11)$ are collinear. Use vectors to find the value of k . [3]

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14. The points A , B , C and D have coordinates $(1, 1)$, $(5, 2)$, $(7, 6)$ and $(3, 5)$.

[4]

(a) Use vectors to show that $ABCD$ is a parallelogram.

(b) Determine whether $ABCD$ is a rhombus, giving a reason.

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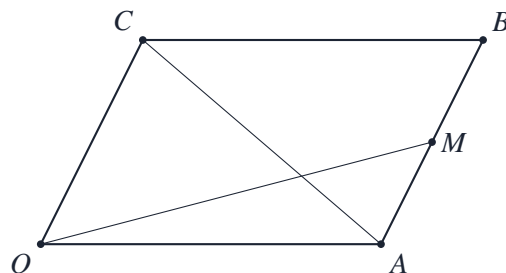
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15. $OABC$ is a parallelogram with $\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.



(a) Use vectors to prove that the diagonals of $OABC$ bisect each other.

[3]

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(b) M is the midpoint of AB . The line OM meets the diagonal AC at the point X .
By writing $\vec{OX} = s\vec{OM}$ and $\vec{AX} = t\vec{AC}$, find the ratio $OX : XM$.

[5]

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(Total for Question 15 is 8 marks)



Section D Vectors in 3D and the scalar product

21 marks

16. Given $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, find $|\mathbf{a}|$ and the unit vector in the direction of $\mathbf{a}$. [2]

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17. The points A and B have coordinates $(1, 0, 3)$ and $(4, -2, 9)$. [3]
Find $\overrightarrow{AB}$ and the distance AB .

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18. Given $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{c} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$, [3]
(a) find $\mathbf{a} \cdot \mathbf{b}$,
(b) show that $\mathbf{a}$ and $\mathbf{c}$ are perpendicular.

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19. Find the angle between the vectors $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, giving your answer to 1 decimal place. [4]

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20. The points A , B and C have coordinates $(2, 1, -1)$, $(5, 3, 1)$ and $(3, -1, 4)$.

(a) Find $\vec{AB}$ and $\vec{AC}$.

[2]

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(b) Find the size of angle BAC , to 1 decimal place.

[3]

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(c) Hence find the area of triangle ABC , to 3 significant figures.

[2]

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(d) The vector $k\mathbf{i} + \mathbf{j} + \mathbf{k}$ is perpendicular to $\vec{AB}$. Find the value of k .

[2]

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(Total for Question 20 is 9 marks)



Section E Vectors in mechanics

20 marks

21. A particle has position vector $\mathbf{r} = (t^2 + 1)\mathbf{i} + 3t\mathbf{j}$ metres at time t seconds. [2]
Find its velocity at $t = 2$.

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22. A particle has velocity $\mathbf{v} = (3t^2 - 2)\mathbf{i} + 4t\mathbf{j}$ m s^{-1} at time t seconds. [3]
Find its acceleration at time t , and its speed when $t = 1$.

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23. A particle has position vector $\mathbf{r} = 2t^3\mathbf{i} + (5 - t^2)\mathbf{j}$ at time t seconds, $t > 0$. [3]
Find the value of t at which the particle is moving parallel to the vector $\mathbf{i} - \mathbf{j}$.

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24. A particle has position vector $\mathbf{r} = (4t - t^2)\mathbf{i} + (t^3 - 3t)\mathbf{j}$ metres at time t seconds. [4]

(a) Find the time at which the particle is moving parallel to $\mathbf{j}$, and its speed at this time.

(b) Find the acceleration of the particle at this time.

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25. A ball is projected from a point O on horizontal ground. Its position vector relative to O after t seconds is

$$\mathbf{r} = 12t\mathbf{i} + (20t - 5t^2)\mathbf{j} \text{ metres,}$$

where $\mathbf{i}$ is horizontal and $\mathbf{j}$ is vertically upwards.

(a) Find the initial speed of the ball. [2]

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(b) Find the time at which the ball reaches its greatest height, and its position vector at this time. [3]

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(c) Find the time at which the ball hits the ground, and its speed at this instant. Comment on your answer. [3]

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(Total for Question 25 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

26. The points A and B have coordinates $(1, 0, 2)$ and $(3, 4, -2)$. The point C lies on the x -axis and angle $ACB = 90^\circ$. [5]
Find the exact possible coordinates of C .

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END OF WORKSHEET

Total: 104 marks